Anthropic Side Challenge Submission

Strategic Integration of Claude 3.5 Haiku

πthons

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Overview: Claude as an Alpha Generator

In the HRT Main Challenge, extracting precise, actionable trading signals from thousands of unstructured financial news headlines was a significant hurdle. We selected **Claude 3.5 Haiku** as our core intelligence engine to bridge the gap between qualitative text and quantitative data.

1. Model Selection & Performance

Processing financial data at scale requires a delicate balance between cognitive depth and speed. We chose Claude 3.5 Haiku because it provided:

- **Massive Throughput:** The ability to process our entire dataset concurrently in record time.
- **Contextual Awareness:** The reasoning capabilities necessary to differentiate between temporary market noise and sustainable trends—a task where basic keyword matching fails.

2. Prompt Engineering & JSON Extraction

We utilized Claude not just as a chatbot, but as a deterministic data-extraction function. By strictly engineering our prompt, we forced the model to return a **strict JSON output**. This allowed for seamless integration into our Pandas and Scikit-Learn pipelines without the need for complex string parsing.

```
# Prompt engineering for deterministic trading signals
prompt = f"""Analyze the impact of this news on the stock price until the end of the trading day.
Respond ONLY with JSON: {{'sentiment': float, 'confidence': float}}
Headlines: {text}"""
```

3. Creative Feature Engineering: Net Conviction

Our most innovative use of Claude was leveraging its self-reflection capabilities. Instead of relying solely on the sentiment score, we extracted Claude's `confidence` metric to create a custom alpha feature called `net_conviction`:

$$\text{Net Conviction} = (\text{Bullish} - \text{Bearish}) \times (1.0 - \text{Uncertainty}) \tag{1}$$

By incorporating this feature, our Machine Learning model (ExtraTrees) learned to apply greater weight to Claude's sentiment signals precisely when the model indicated high confidence in its own analysis.

Cloud-Native Execution

As demonstrated in the attached screenshot, our entire pipeline—including the generation of Claude signals and their integration into our Bad-Tail Risk Filter strategy—was executed flawlessly within our cloud environment.

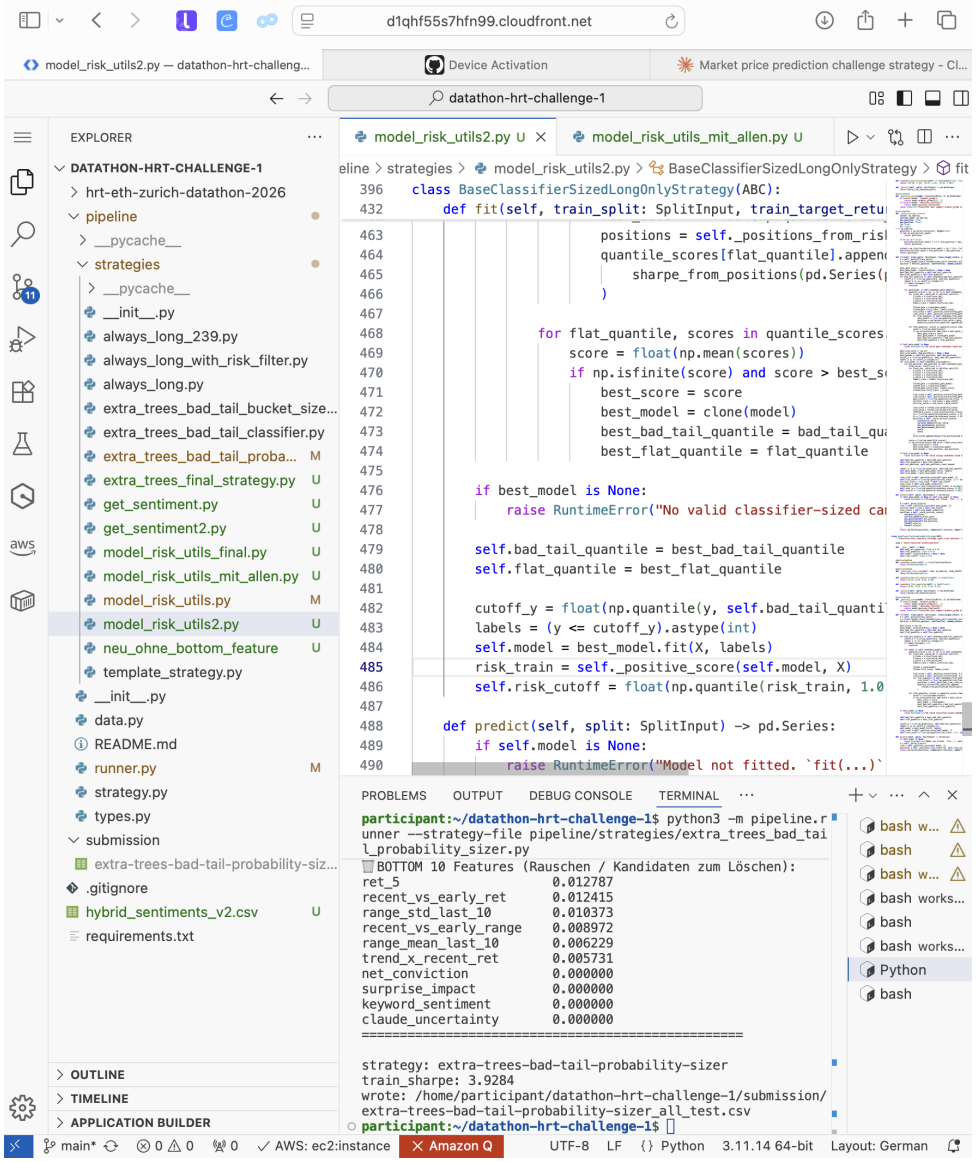

Figure 1: Our cloud development environment demonstrating Claude integration and terminal output.